

Exposé – Master’s Thesis in ICT

Conception and Prototypical Evaluation of a RAG-Based LLM Tutor for Moodle STACK Tasks in Foundational Mathematics Courses

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Abstract

This thesis investigates the design of a RAG-based LLM tutor for STACK tasks in foundational mathematics courses. While STACK continues to perform the mathematical assessment, an external LLM tutor interface generates adaptive didactic hints based on task information, student responses, diagnosis codes, and task-specific RAG contexts. The focus is on how prompt, context, and model selection must be designed so that students are supported during the problem-solving process without being given the complete solution immediately. The thesis includes the development of a minimal prototype as well as its technical and optionally qualitative-didactic evaluation using the think-aloud method.

Contents

Contents	2
1 Background and Problem Statement	3
2 Research Question	5
3 Objectives	6
4 Object of Study and Scope	7
5 Methodological Approach	8
5.1 Requirements Analysis	8
5.2 Selection and Modeling of an Example Task	9
5.3 Development of the Technical Prototype	9
5.4 RAG Concept	10
5.5 Prompt and Tutor Design	10
5.6 Evaluation	11
6 Expected Results and Scientific Contribution	13
7 Sources / Current State of the Art	14
7.1 General LLM-Based Chatbots with Mathematical Capabilities	14
7.2 Mathematics-Specific Tutor and Solver Systems	14
7.3 E-Assessment Systems with Mathematical Assessment	15
7.4 Machine-Learning-Supported Learning Platforms and Tutoring Systems	15
7.5 Positioning within the Current State of Research	15
7.6 Research Gap	16
8 Schedule	17
9 Preliminary Outline	19

◀ 1 Background and Problem Statement

Digital exercise and assessment systems are now well established in foundational mathematics courses at universities. In particular, digital tasks enabled by the open-source e-assessment system **STACK** in Moodle provide symbolic assessment of mathematical input based on the computer algebra system Maxima. This allows not only numerical final results to be assessed, but also algebraically equivalent expressions to be recognized and evaluated. In addition, so-called **Potential Response Trees** can be used to identify typical error patterns and provide specific feedback.

Despite these strengths, feedback in STACK tasks is generally static. It must be formulated in advance by instructors and can only respond to individual solution paths, forms of expression, or comprehension difficulties to a limited extent. At the same time, current Large Language Models open up new possibilities for adaptive, linguistically differentiated, and dialogue-like learning support. LLMs can generate explanations, formulate follow-up questions, and adapt hints to the current learning state. However, their use in mathematical learning environments involves specific risks. These include, in particular, factually incorrect statements, unreliable mathematical assessment, and the direct output of complete solutions.

The planned thesis addresses this interface. Mathematical assessment is to remain the responsibility of STACK, while an external LLM tutor generates didactic hints based on the information provided by STACK. The tutor is intended to use a **Retrieval-Augmented Generation** approach, abbreviated as RAG. Through a task-specific context, the LLM receives relevant information such as the task text, learning objective, typical errors, didactic constraints, hint levels, and, where appropriate, model solutions. The focus is not on the development of a full Moodle plugin, but on the prototypical investigation of a minimal interface for a clearly defined mathematical topic area.

The selected application domain is **differential calculus**, in particular the differentiation of nested exponential functions. An exemplary STACK

◁ 1 Background and Problem Statement

task is:

$$f(x) = -5e^{x^2 - 2e^x}.$$

Solving this task correctly requires, among other things, the application of the chain rule, the differentiation of the inner exponent, and the consideration of the constant factor. Typical errors, such as omitting the inner derivative, can be diagnosed by STACK and passed on to the LLM tutor.

◀ 2 Research Question

The central research question of the thesis is:

How can a RAG-based LLM tutor for Moodle STACK tasks in foundational mathematics courses be designed so that it supports students through adaptive hints without providing the complete solution?

From this overarching research question, the following sub-questions arise:

1. **Information Transfer and System Architecture**
What information must be transferred from a STACK task to an external LLM tutor interface in order to generate task-specific and error-related support?
2. **Prompt and Context Design**
How must the system prompt, task prompt, tutor policy, and RAG context be structured so that an LLM generates mathematically correct, understandable, and didactically appropriate hints?
3. **Prevention of Solution Disclosure**
Which strategies, in particular hint levels, context control, and explicit prompt rules, can reduce the risk that the LLM immediately outputs the complete solution?
4. **Model Selection**
To what extent is a local LLM suitable for this limited mathematical-didactic use case, and how can its performance potentially be compared with that of a commercial model?
5. **Usability and Didactic Quality**
How are the generated hints perceived by students with regard to comprehensibility, usefulness, and support of the problem-solving process?

◀ 3 Objectives

The aim of the thesis is the conception, prototypical implementation, and initial evaluation of a minimal LLM tutor interface for selected STACK tasks from a chosen topic area.

The prototype is intended to realize an information chain from STACK to an LLM-based tutor. STACK continues to perform the mathematical assessment, while the tutor provides didactic support. The central function of the prototype is to load an appropriate RAG context based on a task ID, a student response, and optionally a diagnosis code, generate a structured prompt from this information, and have an adaptive hint generated by an LLM.

Specifically, the thesis pursues the following objectives:

- development of a technical interface for transferring relevant information from STACK to an external LLM tutor,
- creation of a task-specific RAG context for selected differentiation tasks,
- development of a tutor policy with hint levels for controlling didactic support,
- systematic design and testing of prompt variants,
- investigation of strategies for preventing solution disclosure,
- exemplary evaluation of the generated tutor hints,
- derivation of design recommendations for LLM-based tutoring systems in STACK environments.

The thesis is explicitly designed as a prototypical investigation. A complete productive integration into Moodle, comprehensive rights management, or a large-scale empirical effectiveness study are not part of the thesis.

◀ 4 Object of Study and Scope

The object of study is a RAG-based LLM tutor for STACK tasks from a narrowly defined mathematical topic area.

The scope is limited to:

- mathematical topic: differential calculus,
- task type: differentiation of nested functions,
- focus: chain rule for exponential functions,
- technical implementation: external tutor interface,
- didactic focus: adaptive hints without direct solution disclosure.

The following aspects are not considered:

- a complete Moodle plugin,
- a general LLM tutoring platform for arbitrary mathematical topics,
- automatic mathematical assessment by the LLM,
- extensive learning analytics functions,
- a large-scale quantitative study.

This delimitation serves to keep the research question precise and empirically manageable.

5 Methodological Approach

The thesis follows a development-oriented research approach. The focus is on the design and evaluation of the LLM tutor prototype. Methodologically, the project can be described as prototypical software development followed by qualitative and technical evaluation.

5.1 Requirements Analysis

At the beginning, the requirements for the tutor and the information transfer are analyzed. In particular, the following questions are considered:

- What information can STACK provide via a feedback link?
- What information does the LLM need in order to generate a helpful response?
- Which information should deliberately not be provided to the LLM, or only at higher hint levels?
- Which types of errors typically occur in the selected task class?
- Which didactic requirements apply to helpful hints that do not reveal the solution?

The result of this phase is a requirements catalogue for the minimal prototype.

◀ 5 Methodological Approach

5.2 Selection and Modeling of an Example Task

As an exemplary task type, a differentiation task involving the chain rule is selected, for example:

$$f(x) = -5e^{x^2-2e^x}.$$

For this task, typical error patterns are defined, such as:

- missing inner derivative,
- incorrect derivative of e^x ,
- incorrect derivative of x^2 ,
- omitted constant factor,
- sign error,
- differentiation of only the inner exponent.

These error patterns are converted into diagnosis codes that can be passed to the tutor by STACK or by the prototypical interface.

5.3 Development of the Technical Prototype

The technical prototype realizes the minimal information transfer:

$$\text{STACK} \rightarrow \text{Tutor-Interface} \rightarrow \text{LLM} \rightarrow \text{Tutor-Interface}.$$

In the minimal prototype, the response is not returned directly to the STACK interface, but displayed in an external tutor interface opened from STACK. A deeper reintegration into Moodle or STACK can be discussed as a possible future development.

The prototype should provide at least the following functions:

- receiving a task ID,
- receiving the student response,
- receiving an optional diagnosis code,
- loading a suitable RAG context,
- generating a prompt,
- calling an LLM,
- displaying the generated hint.

◀ 5 Methodological Approach

5.4 RAG Concept

The RAG approach is implemented in a deliberately controlled and task-specific manner. Instead of an extensive vector database, file-based retrieval is sufficient in the first step. A corresponding context is loaded via the task ID transferred from STACK.

This context may contain the following components:

- task text,
- mathematical topic,
- learning objective,
- relevant mathematical rule,
- typical errors,
- diagnosis descriptions,
- tutor rules,
- hint levels,
- model solution, where appropriate.

A central aspect is the control of which information is made available to the LLM at which hint level. In particular, it should be investigated whether the model solution should already be included in the context or whether this increases the risk of solution disclosure.

5.5 Prompt and Tutor Design

A major focus of the thesis is the design of suitable prompts. The prompt should clearly position the model as a tutor while simultaneously setting boundaries.

The following aspects must be taken into account:

- the role of the model as a tutor,
- reference to the fact that STACK has performed the mathematical assessment,
- description of the task,
- student response,

◀ 5 Methodological Approach

- diagnosis code or error description,
- RAG context,
- hint level,
- permitted and prohibited content,
- maximum response length,
- request for activating follow-up questions.

Exemplary requirements for the prompt are:

- no independent assessment of the response by the LLM,
- no complete solution at the first hint level,
- reference to the specific error,
- short and understandable wording,
- support for the next thinking or solution step,
- no following of potential instructions contained in the student response.

Different prompt variants can be systematically compared, for example prompts with or without the model solution, with or without diagnosis code, and with different hint levels.

5.6 Evaluation

The evaluation consists of a technical and a didactic-qualitative component.

Technical Evaluation

The technical evaluation examines the functionality of the prototype. The following criteria are considered, among others:

- correct transfer of information from STACK,
- correct loading of the RAG context,
- correct prompt generation,
- successful communication with the LLM,
- correct display of the tutor hint,
- stability of the prototype for typical inputs,
- response times.

◀ 5 Methodological Approach

Didactic-Qualitative Evaluation

The didactic evaluation investigates the quality of the generated hints. Criteria may include:

- mathematical correctness,
- comprehensibility,
- relation to the student response,
- relation to the diagnosis,
- helpfulness for the next solution step,
- prevention of solution disclosure,
- activation of learners through follow-up questions.

Optionally, a small trial with students can be conducted. The **think-aloud method** is suitable for this purpose. Students work on a STACK task, receive a generated hint, and verbalize their thoughts during the process. A short follow-up interview can be conducted as a supplement.

Possible guiding questions are:

- Was the hint understandable?
- Did the hint help identify the error?
- Was the hint too general, appropriate, or too direct?
- Was the solution anticipated?
- Would such a tutor be perceived as helpful during practice?

The qualitative evaluation does not aim to prove general learning effectiveness, but rather to exploratively assess the comprehensibility, usefulness, and didactic appropriateness of the tutor design.

◀ 6 Expected Results and Scientific Contribution

As a result of the thesis, a minimal, functional prototype is expected that realizes the information transfer from STACK to a RAG-based LLM tutor. In addition, prompt templates, a task-specific RAG context, and a tutor policy with hint levels are to be developed.

The scientific contribution lies in particular in the systematic investigation of tutor design. The thesis shows exemplarily how a symbolic-mathematical assessment system and a generative language model can be meaningfully combined without transferring mathematical assessment responsibility to the LLM.

◀ 7 Sources / Current State of the Art

There already exist various commercial systems for mathematical support, AI-supported learning, and digital mathematical assessment. These can roughly be divided into four categories: general LLM-based chatbots with mathematical capabilities, mathematics-specific tutor and solver systems, e-assessment systems with mathematical assessment, and machine-learning-supported learning platforms.

7.1 General LLM-Based Chatbots with Mathematical Capabilities

General LLM-based chatbots include, among others, *ChatGPT*, *Claude*, *Gemini*, *Microsoft Copilot*, and *Perplexity*.

These systems can explain mathematical tasks, formulate solution steps, and respond dialogically to follow-up questions. Their strengths lie primarily in linguistic flexibility and the ability to adapt explanations to different formulations or questions.

For the use case investigated here, however, these systems are only suitable to a limited extent. They do not generally assess mathematical input reliably, often provide complete solutions, and do not have a direct connection to STACK or its Potential Response Trees. In addition, their use is often associated with data protection issues, since processing usually takes place via external cloud infrastructures.

7.2 Mathematics-Specific Tutor and Solver Systems

A second group consists of mathematics-specific systems such as *Photomath*, *Symbolab*, *Mathway*, *WolframAlpha Pro*, and *Microsoft Math Solver*. These applications are specialized in solving mathematical tasks and often provide step-by-step solutions. They are particularly powerful for standard tasks in algebra, analysis, and elementary mathematics.

◀ 7 Sources / Current State of the Art

However, the focus of these systems is predominantly on generating solutions. This fundamentally distinguishes them from a tutoring approach that aims to guide learners toward the next thinking step without immediately providing the complete solution.

The use of structured STACK diagnosis codes or a didactically controlled hint policy is generally not provided in these systems.

7.3 E-Assessment Systems with Mathematical Assessment

Commercial or productively used e-assessment systems with mathematical assessment include, among others, *Möbius* by Maplesoft, *ALEKS* by McGraw Hill, *MyMathLab* by Pearson, *Knewton Alta*, *WirisQuizzes*, as well as non-commercial or open systems such as *STACK* and *WeBWork*. These systems enable the automatic assessment of mathematical answers, partly using symbolic mathematics or adaptive task control.

For the planned approach, it is particularly relevant that these systems provide mathematical assessment and, in some cases, feedback mechanisms, but do not necessarily generate adaptive LLM-based tutorial hints. Feedback is often rule-based, prestructured, or controlled through proprietary mechanisms. An explicit connection between STACK diagnosis codes and a RAG-based LLM tutor is currently not available as an established standard function.

7.4 Machine-Learning-Supported Learning Platforms and Tutoring Systems

Further relevant systems include machine-learning-supported learning platforms and tutoring systems such as *Khanmigo* by Khan Academy, *Carnegie Learning/MATHia*, *Squirrel AI*, *Querium*, and comparable adaptive learning environments. These systems partly pursue more tutorial approaches and attempt to support learners through follow-up questions and adaptive assistance.

Khanmigo is particularly relevant for comparison, as it is designed not merely to provide learners with solutions, but to encourage independent thinking through dialogical support. Nevertheless, it is not a STACK-specific solution.

Direct use of Potential Response Trees, STACK diagnosis codes, or university-specific task metadata is not part of these systems.

7.5 Positioning within the Current State of Research

According to the current state of research, the central challenge in the use of LLMs in mathematical learning environments is not only the generation of

◀ 7 Sources / Current State of the Art

correct solutions, but especially the generation of mathematically correct, didactically appropriate hints that do not reveal the solution. Studies on Large Language Models in educational contexts emphasize risks such as hallucinations, uncontrolled solution output, lack of transparency, unclear responsibility, and data protection issues.

At the same time, research on formative feedback shows that feedback is particularly conducive to learning when it supports the next thinking or action step without replacing the learner’s cognitive activity. From a didactic perspective, the primary objective is therefore not the generation of complete solutions, but the design of adaptive hints that enable learners to continue working independently.

7.6 Research Gap

The planned approach differs from existing commercial systems in that the LLM is not intended to perform the mathematical assessment. Instead, STACK remains the mathematical assessment instance, while the LLM formulates adaptive support based on structured diagnoses and curated contextual information. This coupling of symbolic assessment and generative didactic support constitutes the central object of investigation of the thesis.

8 Schedule

A possible schedule for a twelve-week processing period is shown in Table 8.1.

Table 8.1: Preliminary schedule

Phase	Period	Content
Familiarization and re-finement	Week 1	Review of relevant literature on STACK, LLM tutoring, RAG, and feedback; final delimitation of the topic
Task and error analysis	Week 2	Selection of the example task; analysis of typical error patterns; definition of possible diagnosis codes
Concept development	Week 3	Design of the system architecture; specification of information transfer; definition of the RAG context
Implementation of the prototype	Weeks 4–5	Development of the minimal tutor interface; implementation of the transfer of task ID, response, and diagnosis
RAG and prompt design	Weeks 6–7	Creation of context files; development of prompt variants; definition of hint levels and tutor rules
LLM connection and internal tests	Week 8	Connection of a local LLM; execution of initial test runs; analysis of the generated hints
Optimization of the tutor design	Week 9	Adjustment of prompts; reduction of solution disclosure; optional comparison with a commercial model
Evaluation	Week 10	Technical evaluation; optional implementation of a think-aloud trial with students
Analysis	Week 11	Evaluation of the technical and qualitative results; derivation of design recommendations
Documentation and completion	Week 12	Completion of the written thesis; revision; final preparation for submission

◀ 8 **Schedule**

If a longer processing period is available, the evaluation can be expanded, for example by including additional tasks, several students, or a more systematic model comparison.

◀ 9 Preliminary Outline

1. Introduction

- a) Motivation
- b) Problem Statement
- c) Objectives
- d) Research Question
- e) Structure of the Thesis

2. Theoretical and Technical Foundations

- a) Moodle and STACK
- b) Potential Response Trees and Mathematical Error Diagnoses
- c) Large Language Models in Educational Contexts
- d) Retrieval-Augmented Generation
- e) Didactic Requirements for Feedback and Tutorial Hints

3. Conception of the LLM Tutor

- a) Delimitation to Differential Calculus
- b) System Architecture
- c) Information Transfer between STACK and Tutor Interface
- d) RAG Context and Task Model
- e) Prompt Design
- f) Tutor Policy and Hint Levels
- g) Strategies for Preventing Solution Disclosure

4. Prototypical Implementation

- a) Selection and Implementation of the Example Task
- b) STACK Feedback Link and Parameter Transfer
- c) Tutor Interface

◀ 9 Preliminary Outline

- d) Prompt Generation
- e) LLM Connection
- f) Display of Tutor Hints

5. Evaluation

- a) Evaluation Design
- b) Technical Evaluation
- c) Assessment of Tutor Hints
- d) Optional Think-Aloud Trial
- e) Evaluation Criteria

6. Results and Discussion

- a) Results of the Technical Implementation
- b) Quality of the Generated Hints
- c) Observations on Solution Disclosure
- d) Benefit of the RAG Context
- e) Limitations of the Prototype

7. Conclusion and Outlook

- a) Answering the Research Question
- b) Key Findings
- c) Design Recommendations
- d) Outlook on Moodle Integration and Further Topic Areas